HANYANG ZHONG

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PERSONAL STATEMENT

Robotics PhD candidate specializing in full-stack dexterous manipulation, force-aware teleoperation, and multimodal imitation learning. I build real robotic systems across mechanism design, fingertip force sensing, embedded control, demonstration collection, and policy deployment. My research focuses on reproducible embodied AI systems for contact-rich manipulation.

EDUCATION

The University of York (UoY) *Sep 2026 (expected)* Doctor of Philosophy in Electronic Engineering (Robotics) *Research Areas: Robotic manipulation, Dexterous hand design, Dexterous hand teleoperation, Imitation learning, Force sensing, Embedded systems, SLAM, Human-robot interaction*

The University of York (UoY) *Sep 2022* M.Sc. in Intelligent Robotics *Relevant Courses: SLAM, Robot Kinematics and Manipulation, Machine Learning for Robotics Focus: Robotic perception, sensor fusion, and learning-based manipulation control*

Jiangxi University of Water Resources and Electric Power *Sep 2019* B.Eng. in Electronic Information Engineering *Relevant Courses: Control Systems, Embedded Systems, Digital Electronics, Circuit Design Focus: Electronic hardware design, signal processing, and control algorithms for real-time systems*

PUBLICATIONS AND MANUSCRIPTS

H. Zhong et al., "DexRemo: A Wrist-Frame Fingertip Force Sensing Hand Platform for Contact-Rich Removal Manipulation," **Revise and Resubmit**, *IEEE Transactions on Robotics (T-RO)*, 2026.
Role: Co-first author. Contributions: hardware design, fingertip force-sensor design, and algorithm validation for contact-rich manipulation.

H. Zhong et al., "Hidden Force-Regime Coverage for Contact-Rich Bimanual Imitation Learning," submitted to *Conference on Robot Learning (CoRL)*, 2026.
Role: Independent first author. Contributions: hardware design, full bimanual platform debugging, integration between the robotic system and learning pipeline, sensor design and integration, and algorithm validation.

L. Wang, **H. Zhong**, T. Wang, S. Luo, and J. Zhu, "MLLM-Fabric: Multimodal Large Language Model-Driven Robotic Framework for Fabric Sorting and Selection," accepted for *IEEE Robotics and Automation Letters (RA-L)*, 2025; presented as a poster at *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
Role: Second author and ICRA poster presenter. Contributions: data collection, multimodal large-model fine-tuning, and reasoning-oriented question/prompt design.

H. Zhong et al., "LLM-SAP: Large Language Models Situational Awareness-Based Planning," in *Proc. IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*, 2024, pp. 1–6.
Role: Co-first author.

H. Zhong et al., "FENet: Focusing Enhanced Network for Lane Detection," in *Proc. IEEE International Conference on Multimedia and Expo (ICME)*, 2024, pp. 1–6. **Oral Paper**.
Role: Co-first author.

H. Zhong et al., "Balancing Rigor and Utility: Mitigating Cognitive Biases in Large Language Models for Multiple-Choice Questions," in *Proc. Annual Meeting of the Cognitive Science Society*, vol. 47, 2025.
Role: Co-first author.

WORKING EXPERIENCE

Graduate Teaching Assistant (GTA) *Sep 2022 – Sep 2025* Department of Electronic Engineering, University of York, UK

- Supported MSc Intelligent Robotics teaching across Practical Robotics and Machine Learning, covering ROS-based robot control, sensor integration, model training, debugging, and lab troubleshooting.

- Mentored student teams on robotic system implementation, algorithm-to-hardware deployment, and technical reporting.

Embedded Systems Engineer *Jul 2019 – Aug 2021* Zhigong Control Technology Co., Ltd., Guangzhou, China

- Designed, developed and tested programmable control handles; implemented PCB, firmware, embedded control, hardware debugging, and system integration.

TECHNICAL SKILLS

Robotics: Dexterous manipulation, dexterous hand design, teleoperation, bimanual manipulation, contact-rich manipulation, force-aware manipulation, tactile/force-guided interaction, and real-robot system validation.

Robot Learning: Learning from demonstration, imitation learning, teleoperation-based demonstration collection, ACT-style policy learning, multimodal policy learning, algorithm validation, ablation studies, and policy deployment on physical robots.

Robotic Systems: Python, C++, PyTorch, OpenCV, ROS, Git, Linux-based development, robot control stacks, multi-camera/force/proprioception synchronization, data logging, data processing, and experimental automation.

Hardware and Electronics: Fingertip force sensing, force/torque sensing, sensor calibration, custom PCB design, embedded firmware, motor control, signal acquisition, hardware-software integration, and system-level debugging.

Mechanical Design and Fabrication: SolidWorks, Autodesk Inventor, CAD design, dexterous hand prototyping, force-feedback device design, tactile gripper design, 3D printing, CNC machining, sheet metal fabrication, assembly, and hardware iteration.

Embodied AI and VLA Systems: Multimodal robot perception, VLA-oriented embodied AI systems, LLM-guided robot planning, situational-awareness-based planning, human-robot interaction, and safety-aware decision making.

ACADEMIC SERVICE

- Reviewer, IEEE International Conference on Robotics and Automation (ICRA), 2026.
- Reviewer, Conference on Robot Learning (CoRL), 2026.
- Reviewer, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
- Reviewer, Complex Intelligent Systems (CAIS/CIS), 2026.
- Reviewer, IEEE International Conference on Multimedia and Expo (ICME), 2024.